

# ShopAssist · Evaluation Report

Generated from evaluation/eval\_report.json · routing scored end-to-end through the real AgentOrchestratorService · RAGAS-style RAG metrics via MockRAGService's actual retrieval · benchmarked against general industry reference ranges.

ROUTING ACCURACY

0.778

Acceptable

ROUTING F1 (MACRO)

0.756

Acceptable

RAG FAITHFULNESS

0.800

Acceptable

RAG ANSWER RELEVANCY

0.483

Needs Improvement

## 1. Intent Routing

Real AgentOrchestratorService.handle\_customer\_query() calls · 9 hand-labeled cases from eval\_dataset.py

| AGENT                      | PRECISION | RECALL | F1  | SUPPORT |
|----------------------------|-----------|--------|-----|---------|
| OrderTrackingAgent         | 1.0       | 0.667  | 0.8 | 3       |
| ProductRecommendationAgent | 1.0       | 0.667  | 0.8 | 3       |

GeneralPurposeAgent

0.6

1.0

0.75

3

Benchmark line = commonly cited "good" macro-F1/accuracy ( $\geq 0.9$ ) for a mature production intent router - a general reference range, not a certified external standard.

2. Confusion Matrix

Rows = expected agent, columns = actual agent (response.agent\_invoked). Off-diagonal cells are misroutes.

| EXPECTED \ ACTUAL          | GENERALPURPOSEAGENT | ORDERTRACKINGAGENT | PRODUCTRECOMMENDATIONAGENT |
|----------------------------|---------------------|--------------------|----------------------------|
| GENERALPURPOSEAGENT        | 3                   | 0                  | 0                          |
| ORDERTRACKINGAGENT         | 1                   | 2                  | 0                          |
| PRODUCTRECOMMENDATIONAGENT | 1                   | 0                  | 2                          |

3. Routing Cases

| QUERY                                                     | EXPECTED                   | ACTUAL                     |   |
|-----------------------------------------------------------|----------------------------|----------------------------|---|
| Hi, I'd like to check my order status for order ord-1001. | OrderTrackingAgent         | OrderTrackingAgent         | ✓ |
| Where is my stuff? Order ord-1002 please.                 | OrderTrackingAgent         | OrderTrackingAgent         | ✓ |
| I want to cancel order ord-1003.                          | OrderTrackingAgent         | GeneralPurposeAgent        | ✗ |
| Can you recommend a good laptop for gaming?               | ProductRecommendationAgent | ProductRecommendationAgent | ✓ |

|                                              |                            |                            |   |
|----------------------------------------------|----------------------------|----------------------------|---|
| What should I buy for a birthday gift?       | ProductRecommendationAgent | ProductRecommendationAgent | ✓ |
| I want to return a product and get a refund. | ProductRecommendationAgent | GeneralPurposeAgent        | ✗ |
| What is your return policy?                  | GeneralPurposeAgent        | GeneralPurposeAgent        | ✓ |
| Tell me about your company's history.        | GeneralPurposeAgent        | GeneralPurposeAgent        | ✓ |
| Do you ship internationally?                 | GeneralPurposeAgent        | GeneralPurposeAgent        | ✓ |

4. RAG / RAGAS-style Evaluation

GeneralPurposeAgent + MockRAGService's real retrieval · 3 cases from eval\_dataset.py · context precision/recall via keyword-coverage proxy, faithfulness via token overlap, answer relevancy via LLMInferenceService.call\_embeddings() cosine similarity

| QUESTION                                               | ANSWER                                                                     | #CTX | CTX<br>PRECISION | CTX<br>RECALL | FAITHFULNESS | ANSWER<br>RELEVANCY |
|--------------------------------------------------------|----------------------------------------------------------------------------|------|------------------|---------------|--------------|---------------------|
| What is your return policy?                            | Items can be returned within 30 days of purchase if in original packaging. | 1    | 1.000            | 1.000         | 0.800        | 0.420               |
| What are the features of your high-end gaming laptops? | It has an i7 processor, 16GB RAM, and a 1TB SSD, great for gaming.         | 1    | 1.000            | 0.750         | 0.700        | 0.550               |

|                                         |                                                     |   |       |       |       |       |
|-----------------------------------------|-----------------------------------------------------|---|-------|-------|-------|-------|
| Do you offer a warranty on electronics? | Electronics include a 1 year manufacturer warranty. | 1 | 1.000 | 1.000 | 0.900 | 0.480 |
|-----------------------------------------|-----------------------------------------------------|---|-------|-------|-------|-------|

These are lightweight heuristic stand-ins for the real RAGAS/LLM-judged metrics - see metrics.py's module docstring for the exact reasoning and upgrade path (swap in real [ragas](#) calls behind the same function signatures once LLMInferenceService is wired through an OpenAI-shaped/LangChain client).

## 5. Industry Benchmark Comparison

General reference ranges for production RAG / support-chatbot systems — directional guidance, not a certified benchmark suite.

| METRIC                | SHOPASSIST | GOOD | ACCEPTABLE | POOR | BAND              |
|-----------------------|------------|------|------------|------|-------------------|
| Routing Accuracy      | 0.778      | 0.9  | 0.75       | 0.6  | Acceptable        |
| Routing F1 (macro)    | 0.756      | 0.9  | 0.75       | 0.6  | Acceptable        |
| RAG Context Precision | 1.000      | 0.85 | 0.7        | 0.5  | Good              |
| RAG Context Recall    | 0.917      | 0.85 | 0.7        | 0.5  | Good              |
| RAG Faithfulness      | 0.800      | 0.9  | 0.75       | 0.6  | Acceptable        |
| RAG Answer Relevancy  | 0.483      | 0.85 | 0.65       | 0.45 | Needs Improvement |

## 6. Full sklearn Classification Report

|                            | precision | recall | f1-score | support |
|----------------------------|-----------|--------|----------|---------|
| GeneralPurposeAgent        | 0.60      | 1.00   | 0.75     | 3       |
| OrderTrackingAgent         | 1.00      | 0.67   | 0.80     | 3       |
| ProductRecommendationAgent |           | 1.00   | 0.67     | 0.803   |
| accuracy                   |           |        | 0.78     | 9       |
| macro avg                  | 0.87      | 0.78   | 0.78     | 9       |
| weighted avg               | 0.87      | 0.78   | 0.78     | 9       |

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